TECH BIZ SECURITY · SOC ANALYST INTERNSHIP · BATCH B-503

WEEK 1 · TASK 3

Lab Setup & Windows Fundamentals

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Program	**SOC Analyst Internship (6 weeks)**
Company	**Tech Biz Security**
Date of submission	**21-10-2026**

01 Objective

Before I can investigate an incident I need a safe place to practise and a solid understanding of the system most attacks land on. This week I built my own analysis lab with VirtualBox, running one Ubuntu and one Windows virtual machine, and I completed the TryHackMe room Windows Fundamentals 1. Together they give me a sandbox for the coming weeks and the Windows knowledge a SOC analyst relies on: the file system, permissions, user accounts, UAC, System32 and Task Manager.

02 Proof of Completion

Room / activity	Duration	Tasks	Status
TryHackMe: Windows Fundamentals 1	30 min	10	Completed, room progress 100%
Lab: VirtualBox with Ubuntu and Windows VMs		3 steps	Installed and running (section 03)



Figure 1. Windows Fundamentals 1: all 10 tasks completed (100%).

03 Lab Environment Setup

A SOC analyst should never test suspicious files or tools on a real machine. A virtual machine (VM) is a computer simulated inside my own computer, so anything that goes wrong stays contained and can be undone.

Component	What I set up	Why I need it
Hypervisor	Oracle VirtualBox (Manager window shown in Figure 2).	Runs the virtual machines on my own computer, free of charge.
Ubuntu VM	An Ubuntu Linux desktop, with the Terminal, system monitor and LibreOffice available (Figure 3).	Linux skills and command-line tools are used across security work.
Windows VM	A Windows desktop with the standard taskbar and Edge browser (Figure 4).	Most workstations I will defend run Windows, so I need to know it well.

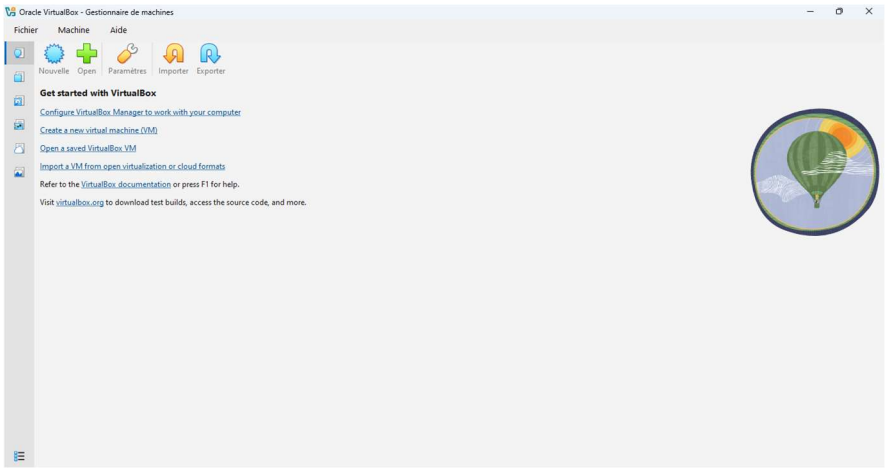


Figure 2. Oracle VirtualBox Manager installed and ready.

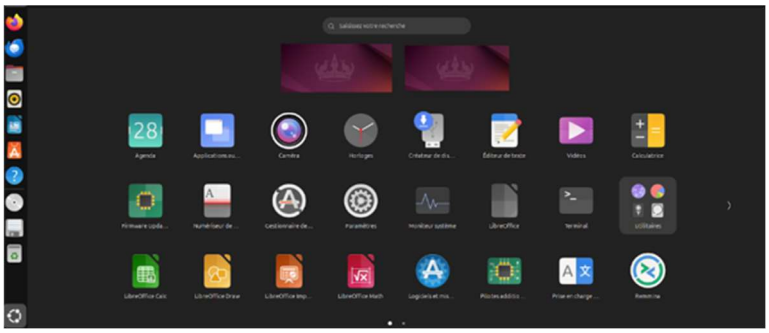


Figure 3. Ubuntu virtual machine: application launcher.



Figure 4. Windows virtual machine: desktop.

04 Windows Fundamentals

4.1 Editions and the desktop

Windows comes in several editions. Home is aimed at individuals, while Pro and Enterprise add the management and security features that businesses need, such as joining a domain and applying group policy. The desktop itself is made of seven parts I learned to recognise:

Desktop element	What it does
Start menu	Opens apps, settings and power options.
Search box (Cortana)	Finds apps, files and settings quickly.
Task View	Shows open windows and lets me switch between virtual desktops.
Taskbar	Holds pinned and running applications.
Toolbars	Optional shortcut bars that can be added to the taskbar.
Notification area	Shows the clock and background programs; useful for noticing things running quietly.
Desktop	The main workspace for icons and files.

4.2 The NTFS file system and permissions

Modern Windows uses NTFS, the New Technology File System. Compared with older file systems it supports files larger than 4 GB, permissions on individual files and folders, compression, and encryption through the Encrypting File System (EFS). Permissions decide who can do what:

Permission	On a folder	On a file
Read	View and list files and subfolders.	View the file's contents.
Write	Add files and subfolders.	Write to the file.
Read & Execute	View, list and run files; inherited by files and folders.	View contents and run the file.
List Folder Contents	View, list and run files; inherited by folders only.	Not applicable.
Modify	Read and write, and delete the folder.	Read and write, and delete the file.
Full Control	Read, write, change and delete everything inside.	Read, write, change and delete the file.

4.3 The System32 folder

The Windows\System32 folder holds the files the operating system cannot work without, and many Windows tools live there. Deleting or changing files in it can make Windows stop working, so I treat it as read-only. For a defender it is also a reference point: a genuine system program normally runs from this folder, and the same name running from somewhere else deserves a closer look.

4.4 User accounts and User Account Control

Account type	What it can do
Administrator	Make system-wide changes: add or delete users, change groups, change settings and install programs.
Standard user	Change only their own files and folders; cannot make system-level changes such as installing software.

User Account Control (UAC) adds a safety step. Even when an administrator is logged in, the session does not run with elevated rights. When an action needs them, Windows asks the user to confirm first. This limits the damage when a user unknowingly starts a malicious program.

4.5 Settings, Control Panel and Task Manager

Settings and the Control Panel are where system options are changed, from networking and user accounts to installed programs. Task Manager shows what is running right now: processes, performance, startup programs, users and services.

05 The Defender's View

Every feature above is something attackers use or abuse, which means each one also shows up in the evidence I will investigate. I have linked them here.

Windows feature	How it can be abused	What I would look for
System32 and system processes	Malware names itself like a system file, for example svchost.exe, but runs from another folder.	A system process whose file path is not C:\Windows\System32.
User-writable folders	Payloads are dropped where any user can write, such as Downloads, Public or ProgramData.	Executables running from those folders.
User accounts	An attacker creates a new local account or adds it to the Administrators group to keep access.	Account-creation and group-change events (Event IDs 4720 and 4732).
Administrator rights and UAC	Malware tries to run with admin rights or bypass UAC prompts.	Unexpected elevation requests or admin-level processes started by a standard user.
Startup programs	Malicious programs are set to start at logon to survive a reboot.	Unknown entries in the Task Manager Startup tab.
NTFS permissions	Weak permissions let a user overwrite files they should only read.	Write access on folders that hold programs.

Quick Task Manager triage checklist

- **Name:** does it look like a real Windows process, or a slightly misspelled copy?
- **Location:** right-click the process and choose Open file location. Is the path where it should be?
- **Publisher:** is the file signed by a known company?
- **Behaviour:** is its CPU, memory or network use unusually high for what it claims to be?
- **Parent and timing:** which program started it, and when?

Task Manager is a good first look. For deeper work, analysts also use tools such as Process Explorer and event logs, which I plan to practise in the coming tasks.

06 Reflection and Key Takeaways

- Knowing what normal Windows looks like, such as the right file paths, accounts and processes, is what lets me recognise abnormal.
- Account, group and service changes are not just admin chores; they are the footprints attackers leave when they try to stay on a machine.
- Together with the networking from Task 2, I now have both halves of the picture: how traffic moves and what it lands on.

Next step: continue with Windows Fundamentals 2 and begin reading Windows event logs in my lab.

07 References

- TryHackMe: Windows Fundamentals 1
- Oracle VirtualBox, Ubuntu and Windows (lab software)
- Microsoft documentation on NTFS and file encryption